ACCESS · INDIGENOUS SKILLS 4 SUCCESS

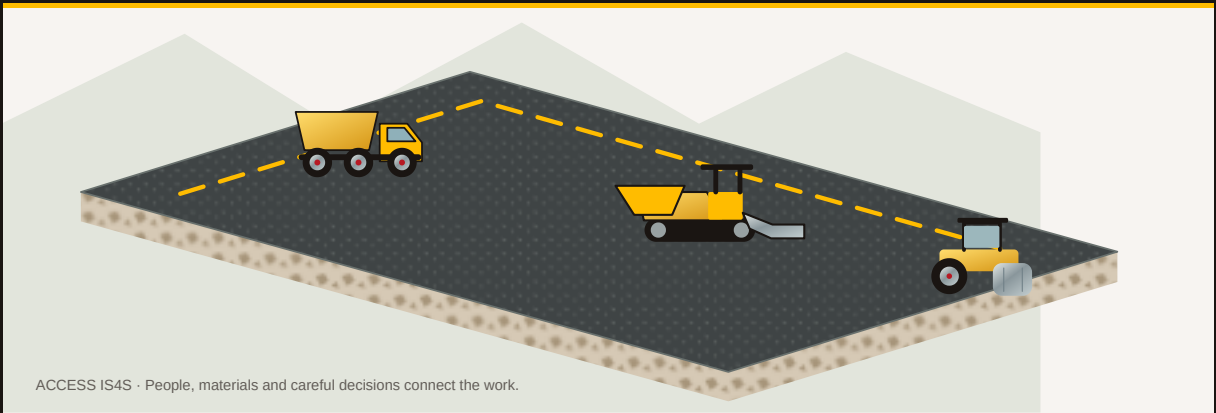
Asphalt paving · Foundations and safe work

Study textbook · Volume 1 · A1–A6 and B1

Classroom development edition · 30 October 2026 · Technical review and pilot pending

ACCESS IS4S / ASPHALT STUDY TEXTBOOK

Roadbuilding connections



ACCESS IS4S. (2026). Roadbuilding connections [Original cover illustration]. Conceptual scene.

Contents

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Asphalt paving · Foundations and safe work

A road is a connection between people

A delivery gets through. A family reaches a service. A worker returns home. Good paving depends on material knowledge, careful preparation and people who understand one another. This volume teaches you to read the work, name what you see, check a quantity and explain a safe next step.

The learning follows **A1–A6 and B1** of the SkilledTradesBC Asphalt Paving/Laydown Technician program outline. Seven chapters contain the readings, definitions, drawings, worked examples and practice information needed for this stage. Later volumes will develop B2–L6. This is the **30 October 2026 classroom development edition**; qualified technical review and classroom pilot are pending. Practical competence requires supervised instruction, observation and the applicable workplace authorization.

FIGURE T01

From rock to a road people use



Note. ACCESS IS4S. (2026). From rock to a road people use [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Your route through the volume

Chapter	You will learn to...	Evidence to keep
1 · The industry and its work	Explain the material route, equipment, road purposes, industry numbers and project documents	A labelled process map and a checked industry brief
2 · How roads changed	Distinguish materials, methods, uses and historical events	A timeline and a careful public caption
3 · Materials, machines and placement	Trace mixture through a plant and paver; identify parts and functions	An annotated machine study and a material-flow explanation
4 · Working in a paving team	Connect roles, responsibilities, communication and learning	A handover and a development plan
5 · Read a project in three views	Read a plan, profile and section; check quantities and a revision	A calculation sheet and map-reading record
6 · Investigate a paving trend	Test a claim against an equivalent service and useful evidence	A comparison brief
7 · Recognize hazards and prepare for safe work	Connect PPE, exposure controls, orientation, WHMIS, first aid, rigging and public protection	A complete classroom safety brief

Use the nine Skills for Success every day

Skill	What it looks like in this book
Reading	Find a condition, symbol or component in the relevant document.
Writing	Leave a clear record that another person can use.
Numeracy	Keep units attached to quantities; estimate and check a result.
Digital	Open, inspect, save and reopen a file or map record on a device.
Problem Solving	Separate a known fact from a missing fact before deciding.
Communication	Explain, listen and confirm shared understanding.
Collaboration	Contribute to a shared task and check another person's work.
Adaptability	Revise a response when the conditions or evidence change.
Creativity and Innovation	Develop and test a clearer explanation or practical improvement.

Keep the activity identifier, your own work, feedback and your revision. A group answer needs a record of each learner's contribution. A paper plan can prepare for a digital task; the digital evidence comes from using the device. A written outcome needs text whose wording you control. Use accessible tools and instructor support as needed, and record that support.

Learn with care for people and place

Road work in B.C. takes place in lands and communities with different histories, rights, relationships and priorities. Ask whose access could change, whose knowledge is needed and who has authority to make a decision. A map cannot supply those relationships. Nobody in the class is expected to represent a Nation, disclose personal experience or share community-held knowledge. Contributions are voluntary; ask before recording or sharing them.

Our Cedar Learning Access case is fictional. Its community facility gives the work a clear purpose without assigning teachings, identities or permissions to a real Nation. Treat classmates' questions as useful information. A learner who identifies uncertainty and seeks the right support is contributing to the crew.

A practical study routine

Look → name → explain → check. Start with the illustration, name the trade feature, explain what it does, then check the caption and the reading. Try the activity before opening the answer guide. Keep an initial attempt and a revision so you can see what changed.

A figure may be an original teaching diagram, a redraw, a manufacturer crop or a screenshot. Its credit identifies the edition and page where relevant. Match a machine's model and serial range before using its operating instructions. General process pictures explain functions; actual equipment procedures and site conditions govern practical work.

Use the worked answers after your attempt. The figure register gives every image's credit and provenance. The interactive plan desk supports Chapter 5 and works offline with this repository.

1 · The industry and its work

SKILLEDTRADESBC · LINE A

A1 · Introduction to the paving industry and paving equipment

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A1.1–A1.5, A1.7 · Industry, equipment, resources, purposes and road types	TB1-01–04; Chapters 2–3
A1.6 · Demonstrate understanding of relevant statistics	LA1-01–02
A1.8 · Review relevant regulations and funding	LA1-03–04
A1.9 · Discuss environmentally friendly	LA1-05–06

Start with the purpose of the road

An asphalt paving/laydown technician helps turn a prepared site and a specified mixture into a pavement that serves its intended use. The work connects materials, machines, people and records. **Infrastructure** means the systems that support daily activity: roads are part of that network, alongside services such as drainage and utilities. A road can carry a delivery, support trade, provide access to school or health care, or serve recreation. The same route may serve several purposes.

The person using the finished road sees the surface. The crew also needs to understand what supports it. A suitable foundation, drainage, appropriate materials and verified construction all contribute to performance. A dark, smooth surface alone cannot tell you whether those conditions were met.

Cedar Learning Access, fictional case. A learning centre needs an access route that will serve people arriving on foot and in vehicles. The owner has appointed a designer and contractor. Before work starts, the crew needs current drawings, an agreed scope, a traffic and public-access arrangement, suitable equipment and a clear safety orientation. A neighbour's request about an entrance matters, but the crew must refer a proposed design change through the authorized project process. Listening and recording the concern helps the right person respond.

Follow the material and the information

Aggregate is mineral material such as sand, gravel or crushed stone. A **pit** is a source associated with excavating deposits such as sand and gravel; a **quarry** typically supplies rock that is extracted and processed. Crushing reduces particle size; screening separates material by size. Suitable material comes from an approved supply and must meet the project requirements. Finding nearby stone does not establish its suitability, extraction permission or environmental approval.

B.C. has varied landscapes and resource locations. Transport distance, the approved grading and properties of the aggregate, access and supply consistency affect a project. The crew uses the specified source and delivery records. A local material can be useful, but “local” is not a quality test. Equally, a long haul can change scheduling and logistics without changing the required product.

Asphalt binder, also called asphalt cement in the handbook, holds mineral particles together in an asphalt mixture. It is different from Portland cement, which is used in concrete. A plant proportions and processes the required materials. A haul vehicle brings the mixture to the paving operation. The paver receives and moves the mixture; its screed contributes to shaping the mat. Rollers compact the placed material, and the work is checked through the required inspection and testing process.

The information travels with the material. A delivery ticket identifies a load; a drawing identifies where the work belongs; a specification defines requirements; a test record reports an observation under a stated method. These records answer different questions. A ticket's mass cannot, by itself, prove the thickness or density achieved in the road.

TB1-01 · Build the material story

Draw five connected stages: source, production, delivery, placement and compaction/checking. At each stage, add one material or machine term, one purpose and one record or question. Explain the route to a partner using *aggregate*, *binder*, *paver*, *screed* and *compaction*. Your partner points to the stage being described, then you swap. Each learner keeps an independently labelled map. **Skills:** Reading, Communication, Collaboration, Creativity and Innovation.

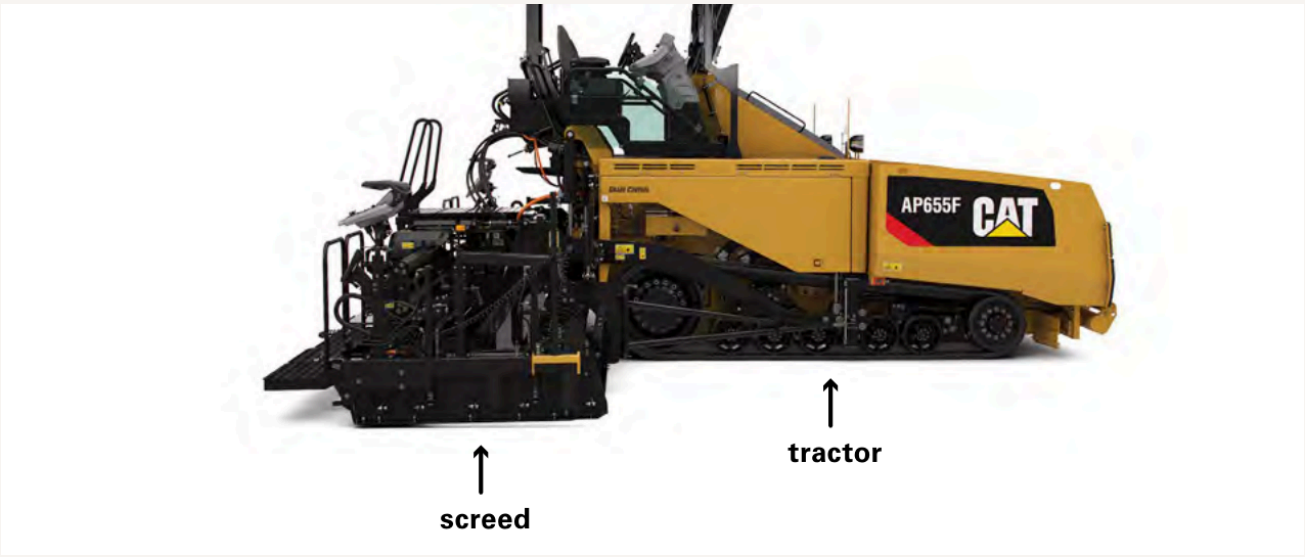
Name the equipment by what it does

Equipment or component	Main function in this introduction	A distinction to remember
Crusher and screen	Reduce and separate aggregate sizes in an appropriate processing operation	Processing equipment does not establish that the product meets a specification.
Asphalt plant	Produce the specified mixture through the applicable production arrangement	Batch and continuous processes differ; Chapter 3 follows both.
Haul truck	Transport material and support controlled delivery	A load ticket is a record, not a placement test.
Paver tractor	Provide propulsion, power and material-handling functions	The tractor and screed are the two main sections shown below.
Hopper and conveyors	Receive mixture and move it rearward through the paver	The conveying mechanism has moving parts and hazardous energy.
Augers	Distribute mixture across the working width ahead of the screed	Their motion differs from the rearward conveyor route.
Screed	Shape the mat as part of the paving system	It responds to several interacting factors; it is not a fixed-depth stamp.

Equipment or component	Main function in this introduction	A distinction to remember
Roller	Compact material through the applicable procedure	A smooth appearance or a pass count alone does not prove density.
Rake, lute and other hand tools	Support appropriate spreading, shaping and finishing tasks	Similar-looking tools have different heads, functions and limits.

FIGURE T02

Two main sections of the paver

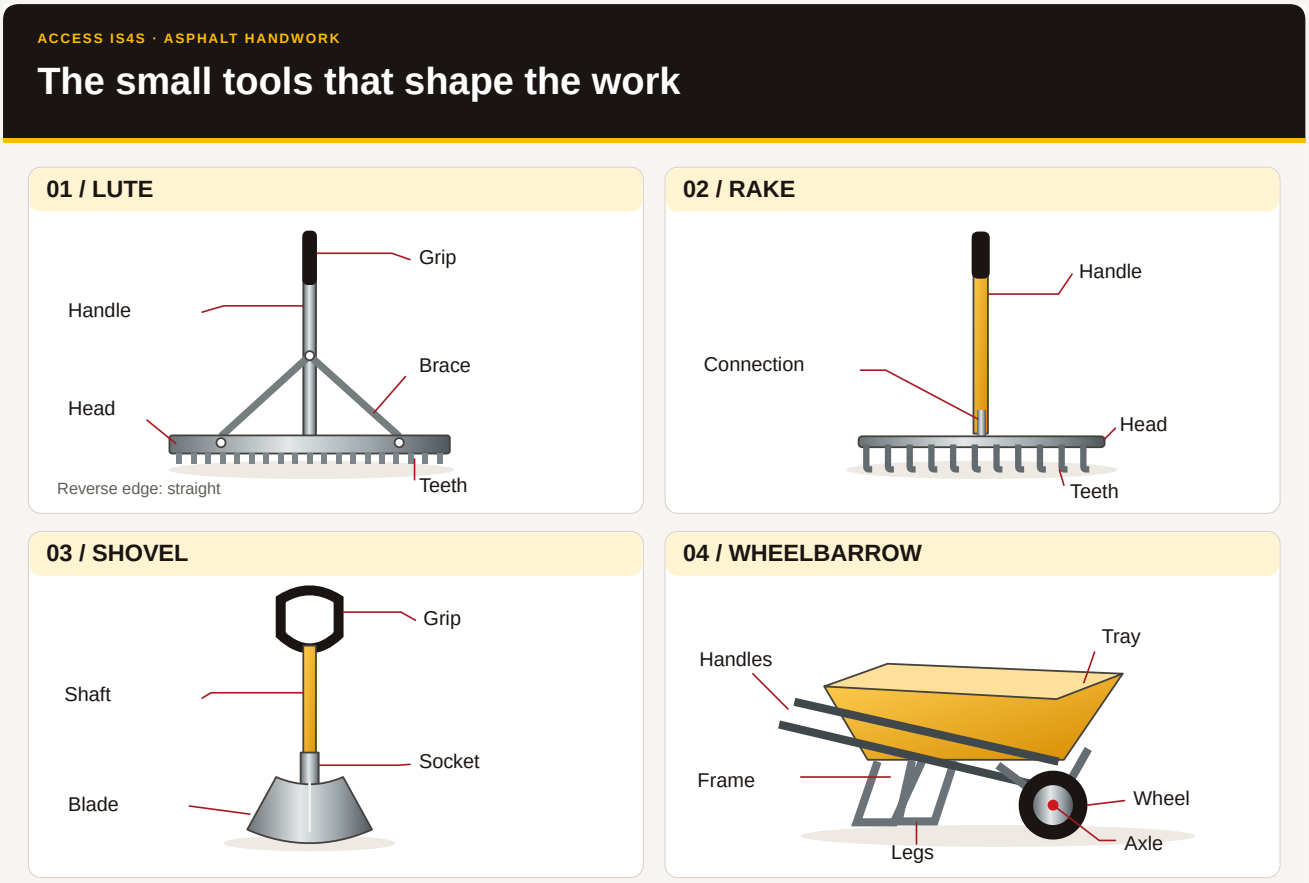


Note. From Guide to Asphalt Paving (QEBQ1888, source PDF p. 63), by Caterpillar, 2016. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Conceptual paver example in the 2016 application guide; this picture does not identify a specific machine serial number. Source and context.

Look and explain. Locate the screed in the manufacturer picture. Then locate the tractor. Explain how a machine can have one section that moves and handles material and another that contributes to shaping the mat. These are functional categories; an actual machine may have additional systems and different controls.

FIGURE T03

Name the parts of the handwork kit



Note. ACCESS IS4S. (2026). The small tools that shape the work [Original instructional SVG]. Illustrative anatomy; confirm the actual model and parts list. Source and context.

A **lute** commonly has a broad head used in asphalt handwork. A **rake** has teeth suited to its intended material-handling task. Product terminology and head arrangements vary. Use the actual product description and instructions to distinguish a toothed head, a straight edge, a brace and a handle connection. A **shovel** has a blade and shaft, often with a grip; a **wheelbarrow** has a tray, handles, frame, wheel arrangement and supports. Loose connections, damaged handles, sharp damage or an unsuitable load can change the risk even when the tool looks familiar.

FIGURE T04

Match the drawing to real tools



Note. From Partner Resource Guide (source PDF p. 156; printed p. 155), by Seymour Midwest, 2026 publisher download. Copyright Seymour Midwest. Upper illustration area cropped; original product relationships retained. Source and context.

TB1-02 · Connect a part to a purpose

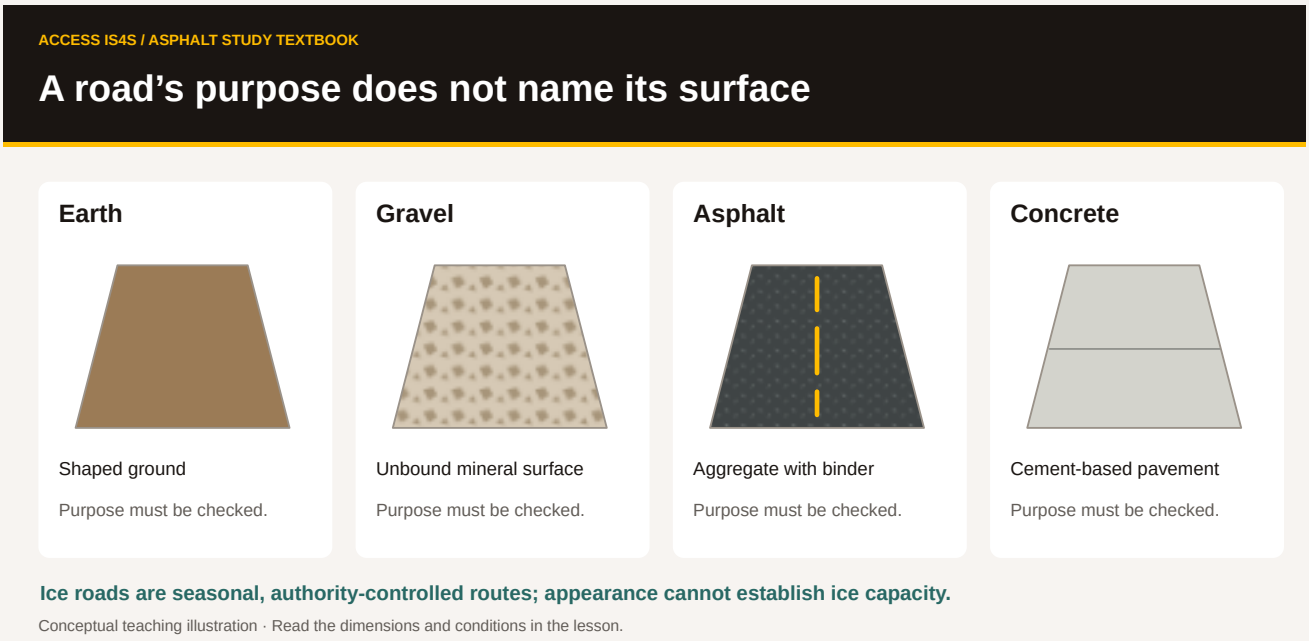
Identify tractor and screed in the CAT image. On the hand-tool diagram, identify a lute head, brace and handle; a rake's teeth; a shovel blade; and a wheelbarrow frame and wheel. Compare one diagram with the manufacturer's catalogue photograph. Record one feature that agrees and one feature that must be checked on the actual tool. Write a short defect report for the fictional observation “the rake head moves at its handle connection.” Describe what you observed and who should assess it before use; do not invent a repair. **Skills:** Reading, Writing, Problem Solving.

Roads have different surfaces and purposes

An **earth road** uses the ground surface, which may be shaped or improved. A **gravel road** has an unbound mineral surface. **Asphalt pavement** uses an asphalt-bound surface; **concrete pavement** uses a cement-based material. Supporting layers and drainage vary with the design. An **ice road** is a seasonal route whose opening, loading and use require the relevant authority's controls. A photograph does not establish safe ice capacity or permission to travel.

FIGURE T05

A road’s purpose does not name its surface



Note. ACCESS IS4S. (2026). A road’s purpose does not name its surface [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

A **resource road** describes an access context, often connected with resource activity; it does not name one surface. A **logging road** describes a use. A **corduroy road** describes a construction form using logs laid across the direction of travel. Those three terms are not interchangeable. Chapter 2 develops the history and the evidence for different methods.

TB1-03 · Classify carefully

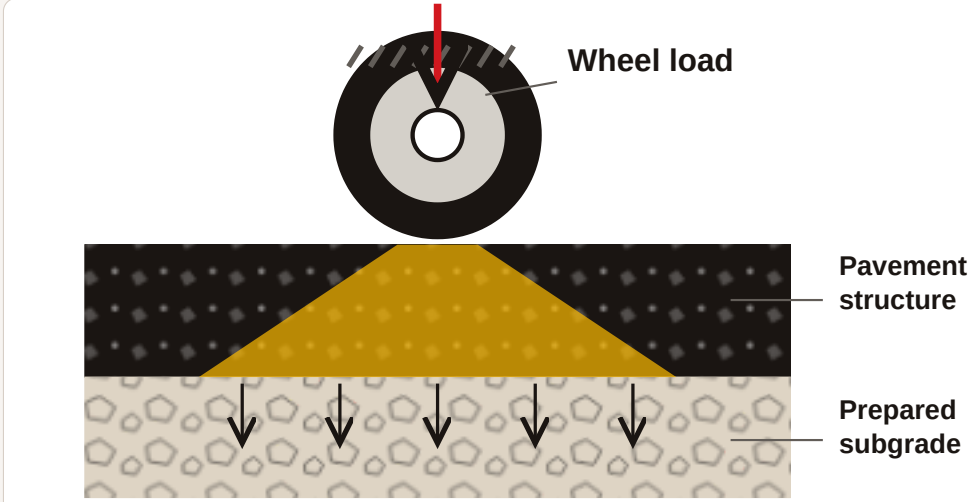
For each phrase, identify whether it primarily describes material/surface, construction method or use: gravel road; logging road; corduroy road; concrete pavement; recreational access; seasonal ice road. Some phrases carry more than one kind of information—explain your choice. Then write three ways a road can support a community and two questions about effects on people, land or water that would need project-specific discussion. Keep personal or community-held knowledge private unless its sharing is agreed. **Skills:** Reading, Writing, Problem Solving.

Explain why the foundation matters

A pavement carries loads through its structure to the prepared ground. The layers work as a system. Their materials, thicknesses and condition influence how the system responds. The illustration below shows a qualitative idea: the region carrying the load becomes wider through the pavement structure. It is not a pressure calculation or a design method.

FIGURE T06

Spread the load



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 1.5, p. 13 / PDF 47. Simplified qualitative section; replaced pressure profile with a shaded load-transfer area and plain-language labels. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

TB1-04 · Give a complete first explanation

Prepare a one-minute explanation for a new classmate: what asphalt paving is, why roads matter, how aggregate and binder reach the road, which equipment functions you can now identify, and why supporting layers need attention. Include one thing the pictures cannot establish about an actual job. Ask the listener to restate your explanation; revise a sentence that was unclear. Save and reopen a typed or recorded version on a device, with permission before recording another person. **Skills:** Communication, Adaptability, Digital.

Read a number with its label

A statement about an industry needs a measure, geography and period. A count of people employed is different from a count of employers. A forecast of openings over ten years is different from the number employed in one year. Construction includes building work and civil engineering as well as paving. A broad construction figure provides context; it cannot be renamed an asphalt-paving total.

Evidence card I1 · B.C. construction. WorkBC's Construction industry profile, retrieved 29 October 2026, reports the following. The 2024 values are attributed on the page to the Labour Force Survey. The openings forecast is attributed to the B.C. Labour Market Outlook.

Measure	Period	Reported value	Population or scope
Employment	2024	246,750	B.C. construction industry
Full-time share	2024	92%	People employed in B.C. construction
Part-time share	2024	8%	People employed in B.C. construction
Full-time share	2024	80%	People employed across B.C. industries
Part-time share	2024	20%	People employed across B.C. industries
Forecast job openings	2025–2035	87,280	B.C. construction industry

The public page does not supply an asphalt-specific employer count in this extract. Shares are published rounded values; multiplying them by an employment total produces an approximation, not a newly observed headcount. Forecasts describe expected demand under assumptions and do not guarantee a job or a person's hours.

Evidence card I1b · Road-industry business locations. Statistics Canada's table 33-10-1095-01, *Canadian Business Counts, with employees, December 2025*, released 13 February 2026, reports **427 active statistical locations with employees** in B.C. under **NAICS 237310, Highway, street and bridge construction**. This is a closer road-industry classification than all construction, but it includes more than asphalt paving. A company with several statistical locations can be counted several times. Locations are not a count of unique employers, people or projects.

Employment-size category	Reported B.C. statistical locations
1–4 employees	175
5–9 employees	63
10–19 employees	72
20–49 employees	62
50–99 employees	32
100–199 employees	18
200–499 employees	4

Employment-size category	Reported B.C. statistical locations
500 or more employees	1
Total with employees	427

The table's notes explicitly say not to use the size ranges to calculate total employment, and advise against using these counts as a time series because methodological changes can affect differences. The retained machine record has reference-period code `2025-07`; the publisher labels the semi-annual table **December 2025**. Keep both in a source record rather than silently rewriting the downloaded field.

Worked example. The difference between the construction full-time share and the all-industry share is `92% − 80% = 12 percentage points`. It is not a 12% increase in employment. An approximate construction part-time count would be `246,750 × 0.08 = 19,740 people`, using a rounded share. A careful sentence keeps the word *approximately* and the 2024 construction scope.

LA1-01 · Report what the data can support

Write a four-sentence industry note using I1: one employment statement; the full-time-share comparison; one forecast statement; and one limitation. Correct this claim: “There are 246,750 asphalt employers in B.C. and 87,280 guaranteed new paving jobs.” Then use I1b to check the sum of the eight size categories, find the difference between the 1–4 and 5–9 categories, and find the 1–4 category's share of all 427 locations to two decimal places. Explain why neither adding category midpoints nor subtracting 427 from 246,750 gives a valid workforce result. Mark the publisher, period, measure and scope beside each number. **Skills:** Reading, Numeracy, Writing.

LA1-02 · Retrieve and preserve a source record

Using the public WorkBC link in the research record, locate one I1 value on a device. Record the page title, publisher, address, access date, measure and reporting year. Compare it with I1; a changed page does not make the retained dated extract disappear. Save and reopen your note. If access is unavailable, use the printed card for reading and schedule a later digital turn. **Skills:** Digital, Reading, Adaptability.

Separate law, contract information and funding

Evidence card I2 · Workplace orientation. B.C.'s Occupational Health and Safety Regulation, sections 3.22–3.25, defines a new worker to include someone new to a workplace and people affected by changed workplace hazards. Section 3.23 requires the employer to provide workplace-specific health and safety orientation and training before a young or new worker begins work. Topics include supervisor contact information, rights and responsibilities, hazards, emergency procedures, and instruction and demonstration of the work task. Section 3.24 addresses additional training when observation shows it is needed or when requested; section 3.25 requires records. This card identifies selected provisions; it is not the complete orientation.

Evidence card I3 · Highway specifications. B.C.'s Standard Specifications for Highway Construction page, updated 1 April 2026, describes the specifications as a standard reference in construction contracts. It identifies the 2025 edition as adopted 1 February 2025 and an amendments appendix updated 1 April

2026. The page says modifications for local conditions are specified in special provisions for each contract. A current public edition is a starting point: the signed contract, drawings, amendments and special provisions establish which requirements apply to a particular job. A specification reference is not automatically a statute.

Evidence card I3b · A defined abbreviation. The 2006 *Design Build Standard Specifications for Highway Construction*, clause **502.03.01**, defines **EPS as End Product Specification**. In that clause, the contractor is responsible for construction-process quality control and the Ministry performs specified end-product audit sampling and testing for acceptance/rejection and payment adjustment. The definition is confirmed in PDF page 297, printed 1 of 22. That historical definition does not establish which edition governs a current contract, or prove the intended meaning of every unexplained use of EPS. In a geofoam discussion, EPS means expanded polystyrene; read the surrounding subject and definition.

Evidence card I4 · A completed project. The Kicking Horse Canyon Phase 4 project pages report completion on 31 July 2024. The project overview lists Government of Canada investment of **$215.19 million** and B.C. government investment of **$385.58 million**. The procurement page records the Province awarding the design-build contract to **Kicking Horse Canyon Constructors** on 6 November 2020. That team consisted of Aecon Group Inc., Parsons Inc. and Emil Anderson Construction. Funding is not the same as a contractor's profit or an individual worker's pay. The public summary is not the project's complete contract.

LA1-03 · Match a question to a document

For each question, choose I2, I3, I3b, I4 or “additional document needed,” and explain the limit: What training does a new worker need before starting? Which specification edition governs a particular signed contract? Who are the two reported public funders? What is the contractor's profit? What does EPS mean in clause 502.03.01 of the 2006 specifications? What further context is needed to interpret EPS in an old note that gives no definition? Create a document register with title, issuer, date/version, status and question still open. **Skills:** Reading, Problem Solving, Writing.

LA1-04 · Check the funding statement

Add the two I4 contributions and calculate the federal share of that sum to two decimal places. Explain why “more than $600 million” is consistent with the listed values. Rehearse a 45-second explanation distinguishing funder, contracting authority and contractor. Your partner checks the roles and units, then you swap. **Skills:** Numeracy, Communication, Collaboration.

Compare environmental claims within a boundary

Asphalt and concrete are pavement materials, not complete environmental outcomes. Compare a defined service, time period and system boundary. A material-production figure alone leaves transport, construction, maintenance, use and end-of-life outside the answer. A reusable material is not automatically the option with the lowest total impact. The amount recovered, replacement of new material, processing energy, transport and required performance all matter.

Evidence card I5 · Material recovery, a bounded comparison. The U.S. EPA's March 2004 *Hot Mix Asphalt Plants* chapter describes asphalt mixtures that may include reclaimed asphalt pavement, or RAP, alongside aggregate and asphalt cement. FHWA's undated historical *Recycled Concrete Aggregate* review describes concrete being recycled as aggregate, principally for base material in the review's U.S. context. These are two evidenced recovery pathways. Neither card provides a matched B.C. life-cycle emissions comparison, equal pavement designs, haul distances or a common service life. They support "both materials can enter recovery pathways," not "either material is always greener." The references describe mechanisms rather than current local market shares.

Practice decision brief, fictional. A community facility is considering two pavement options for the same 50 m by 6 m access route over a proposed 20-year comparison period. Both must meet the same approved function and loading requirements, which a qualified designer would establish. Option A proposes asphalt with some RAP; Option B proposes concrete with some recycled aggregate. The proposer supplies no verified quantities, maintenance plan, emissions factors or transport distances. The group must decide whether a claim of "30% lower whole-life emissions" has been established. No numerical environmental advantage is supplied as fact.

LA1-05 · Make the strongest supported comparison

Create a comparison grid with material production, transport, construction, maintenance/use and end-of-life rows. For each option, distinguish the recovery pathway supported by I5 from information missing in the fictional brief. Write a conclusion of 60–90 words. Include the common function, 20-year proposed period, one supported similarity and at least three data needs. **Skills:** Reading, Problem Solving, Writing.

LA1-06 · Revise a claim when the scope changes

A proposer now says the "30%" applies only to factory production per tonne and compares different product quantities. Explain why that does not answer the whole-route, 20-year question. In pairs, redesign the comparison request so the next submission could be evaluated. Each learner records one contribution and one revised sentence. **Skills:** Numeracy, Collaboration, Adaptability, Creativity and Innovation.

Keep your industry note, source record, document register, funding calculation and comparison request. An instructor checks the evidence and the questions you leave open, not whether you force a confident answer from incomplete information.

2 · How roads changed

SKILLEDTRADESBC · LINE A

A2 · Describe history of asphalt pavement

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A2.1 · Describe evolution of roads	LA2-01, LA2-06
A2.2 · Describe products used	LA2-02, LA2-05
A2.3 · Describe logging roads	LA2-02–03, LA2-05
A2.4 · Explain methods of placement	LA2-05
A2.5 · Rationale for more roads	LA2-04, LA2-06

Ask what a road was built to do

Road history is a history of materials, movement, power and relationships with place. A route can serve travel, trade, access to resources, settlement or military movement. Its purpose can change. A road that enables one activity may also affect land, water, access and existing travel routes. A timeline should identify whose account is being used and avoid treating one engineering tradition as the beginning of all movement.

In B.C., a classroom discussion of road history does not give permission to map or publish community-held routes or knowledge. Use the public evidence cards below. Invite local knowledge only through the appropriate relationships and agreed purpose. The lesson asks you to explain change, not to speak for a community.

Say what kind of word it is

Trade word	Everyday explanation	Avoid this mix-up
Corduroy road	A construction form using closely laid logs across the direction of travel	The pattern describes a method; it does not prove a logging purpose
Logging road	A road used in forestry access or timber movement	It can have different surfaces and construction methods
Placement	Putting the prepared material into its intended position and form	Delivery gets it to the work; placement puts it in the layer

Trade word	Everyday explanation	Avoid this mix-up
Compaction	Making the placed material denser through the applicable process	A smooth-looking surface alone does not prove specified density

Try the words aloud. “Timber is the material; corduroy is the construction form; logging is one possible purpose.” Use this three-part pattern in LA2-02.

Evidence cards for a timeline

H1 · Via Appia. UNESCO describes the Appian Way as a major Roman road constructed and developed from **312 BCE to the fourth century CE**. Its initial strategic purpose included military conquest. It later supported connected settlements, agriculture and trade. Engineering works included the road and associated structures. One road can have several purposes over time; “Roman” does not mean “the world's first road.”

H2 · Alaska Highway. Public Services and Procurement Canada's history, updated **16 December 2025**, explains that the U.S. Army Corps of Engineers built the road during the Second World War to provide land access to Alaska. It identifies local First Nations' vital role as guides. The pioneer road was completed in **1942**, and the permanent location in **1943**. Responsibility for the Canadian portion transferred to National Defence in **1946**. Official civilian non-military use began in **1948**. Further responsibility changes occurred in **1961**, **1964** and **1992**. These are different events, not competing dates for the same event.

H3 · Post-war highways. FHWA identifies the U.S. Federal-Aid Highway Act of **1956** as a major step in the Interstate system. Its historical discussion connects the system with national planning and everyday movement of goods and people. The 1950s were within the Cold War period. A date and a national-security context do not prove that every new road had one motive; use evidence for the particular project.

H4 · Present resource roads. B.C.'s resource-road information describes a network used for resource industries and access. A resource road is not necessarily paved, and a logging road is not necessarily a corduroy road. Its owner, administrator, condition and access rules need to be established from the relevant record. A present-day category does not describe every historical method used to build it.

FIGURE T07

Dates tell different stories



Note. ACCESS IS4S. (2026). Dates tell different stories [Original instructional SVG]. Based on the public historical accounts by Public Services and Procurement Canada (2025), UNESCO (2024) and FHWA listed in the references. Separate time scales; no universal sequence implied.

LA2-01 · Build a timeline that preserves meaning

Place 312 BCE, 1942, 1943, 1946, 1948 and 1956 in chronological order. Label each event and its evidence card. Identify two different meanings of “completed” in H2. Write one sentence explaining why a timeline alone cannot show every social or environmental effect of roadbuilding. **Skills:** Reading, Writing, Problem Solving.

Materials and methods solve different problems

An earth route uses the ground surface, sometimes shaped or improved. Stone and gravel add mineral material. Cut or shaped stone can form a durable surface where the design, foundation and maintenance suit the use. Wood has also been used: a **corduroy road** is a historical construction form using closely laid logs across the travel direction, often associated with soft or wet ground. The log pattern explains the name. It is a method, while *logging road* describes a road's use; the categories overlap in some histories but are not synonyms.

Asphalt pavement combines aggregate with a bituminous binder. Concrete pavement uses a cement-based binding system. These materials have different production and construction processes. Do not confuse asphalt cement—the binder in an asphalt mixture—with Portland cement used in concrete. A road may have several layers and may be repaired with a different material from its original surface. Appearance alone is not enough to identify its entire construction.

Modern asphalt placement commonly involves delivery of mixture, controlled placement through a paver and screed, and compaction using rollers, with associated preparation, checks and finishing. Mechanization changes how material is moved and shaped; it does not remove the need for a suitable foundation, drainage, specifications or skilled people. This unit describes the change in method. Operating settings belong to the relevant supervised training and approved documents.

Worked example · Correct a category mistake. A museum draft says, “All logging roads were corduroy roads, and modern roads replaced wood with asphalt.” The first clause confuses purpose with method. The second implies one universal sequence. A better label is: “Corduroy construction used logs across a route. It was one historical way of addressing difficult ground. Logging roads have used varied surfaces and construction methods; modern asphalt paving is one distinct technology.”

LA2-02 · Match material, method and purpose

Create three columns headed material, method and purpose. Sort these terms: earth; cut stone; gravel; timber logs; concrete; moving harvested timber; military access; asphalt mixture; closely laid crosswise logs; delivery/placement/compaction; travel to a community facility. Explain the difference between asphalt cement and Portland cement. Explain why a purpose can be served by more than one material. Add one question you would ask before classifying an unfamiliar road from a photograph. **Skills:** Reading, Communication.

LA2-03 · Explain a road without inventing its history

Fictional archive label: “Photo R17, date not recorded. A line of logs crosses a wet-looking route. A truck appears in the background. No owner, purpose or place is recorded.” Write a 50–80-word description separating observation, reasonable interpretation and unknown information. A partner reads it and identifies any unsupported claim. Revise one sentence. **Skills:** Writing, Collaboration, Adaptability.

Explain why demand changed

War can accelerate road construction, as H2 shows. Post-war planning and Cold War concerns formed part of the setting for major highway development, while trade, industry, settlement and daily access also influenced demand. The same road can support freight, recreation and local travel. “More roads” is not a complete measure of benefit: maintenance, alternatives, land use and the people affected belong in the question.

LA2-04 · Compare two explanations

Compare these claims: “The Alaska Highway began as a wartime access project” and “All highways were built only for war.” Use H2 and H3 to support, limit or reject each claim. Calculate the interval from the 1942 pioneer-road completion to the 1948 opening to civilian non-military use. Explain why that interval does not mean no work occurred between those dates. **Skills:** Reading, Numeracy, Problem Solving.

LA2-05 · Connect a historical method to a modern sequence

Make a two-panel classroom display: one panel explains a corduroy route using the supplied definition; the other explains modern asphalt delivery, placement and compaction. Label material, method, purpose and one unknown in each. Use words, a simple sketch or an accessible digital layout. Do not provide a construction recipe. Each partner explains one panel and checks the other's labels. **Skills:** Creativity and Innovation, Collaboration, Communication; Digital if a tool is actually used.

LA2-06 · Independent transfer: a careful public caption

Write a 90–120-word caption titled “Roads change as needs and methods change.” Include a dated example, a material/method distinction, a reason for road growth and a boundary on what the evidence establishes. Add a source note from the evidence register. Keep the first and revised versions. Your next step may be a local public-archive enquiry, but record it as a proposed enquiry until evidence is actually obtained.

3 · From materials to paving

SKILLEDTRADESBC · LINE A

A3 · Describe paving technology

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A3.1 · Describe two types of asphalt plants	LA3-02
A3.2 · Describe asphalt ingredients	LA3-01, LA3-05
A3.3 · Discuss required resources	LA3-01
A3.4 · Describe liquids used in asphalt production	LA3-03
A3.5 · Describe asphalt recycling technology	LA3-04, LA3-06
A3.6 · Describe the asphalt “Train”	LA3-04, LA3-06

Follow the material through the system

Paving technology connects material sources, manufacturing, transport and placement. Begin with the function of each stage. A pit or quarry supplies mineral material; processing prepares aggregate to specified characteristics; a plant combines selected materials; haul vehicles move the mixture; paving equipment places and shapes it; rollers compact the placed layer. Preparation, testing, traffic arrangements and finishing also matter. A flow map is a way to understand relationships, not permission to operate equipment.

Figure LA-1

Follow the material, step by step



Note. An original ACCESS IS4S instructional illustration. The process distinctions are explained in this chapter; the drawing supplies no operating settings or mix proportions. By ACCESS IS4S (2026a); see the figure references in the working reference section.

Read the trade word, then explain its job

Trade word	Plain-language meaning	Say it in context
Asphalt cement / asphalt binder	The binding material in an asphalt mixture	“The binder holds the aggregate together.”
Processed gravel / aggregate	Mineral material prepared for the required characteristics	“The aggregate source still needs to meet the specification.”
Batch plant	Makes identifiable batches of mixture	“This record identifies one batch.”
Drum-mix plant	Uses a continuous material and mixing process; layouts vary	“This record describes a continuous feed period.”

Trade word	Plain-language meaning	Say it in context
Pit / quarry	Sources of mineral material: unconsolidated deposits / rock	“The source name does not prove material suitability.”
Asphalt train	A coordinated set of equipment and roles	“Name the process before explaining the train.”

The outline uses **asphalt cement**. This book also uses **asphalt binder** for that material. Keep both terms available when reading different documents; neither means Portland cement. Use the complete phrase when the short word *cement* could be misunderstood.

Know what the ingredients do

Aggregate is mineral material such as processed sand, gravel or crushed stone. A **pit** generally obtains unconsolidated deposits; a **quarry** extracts rock that may need crushing and screening. Material availability does not establish suitability: the required properties come from the approved design and specification. A short haul distance alone cannot answer whether a source meets the project requirements.

Asphalt binder, also called asphalt cement in many documents, binds the aggregate in asphalt mixtures. It is distinct from the Portland cement used in concrete. **Additives** and modified binders can change mixture characteristics for a defined purpose, but their selection and proportions require the applicable approved information. Fuel used to heat a production process is not automatically an ingredient in the pavement mixture. A delivery label must identify the actual product rather than relying on the loose word *oil*.

Technology evidence card T1. EPA's *Hot Mix Asphalt Plants* description identifies sized aggregate, liquid asphalt cement and potentially reclaimed asphalt pavement as components of hot-mix asphalt. It describes batch and drum-mix production. Its technical account is used here for the process distinction; its historical U.S. production totals and operating details are not current B.C. specifications.

LA3-01 · Explain the ingredients to a new learner

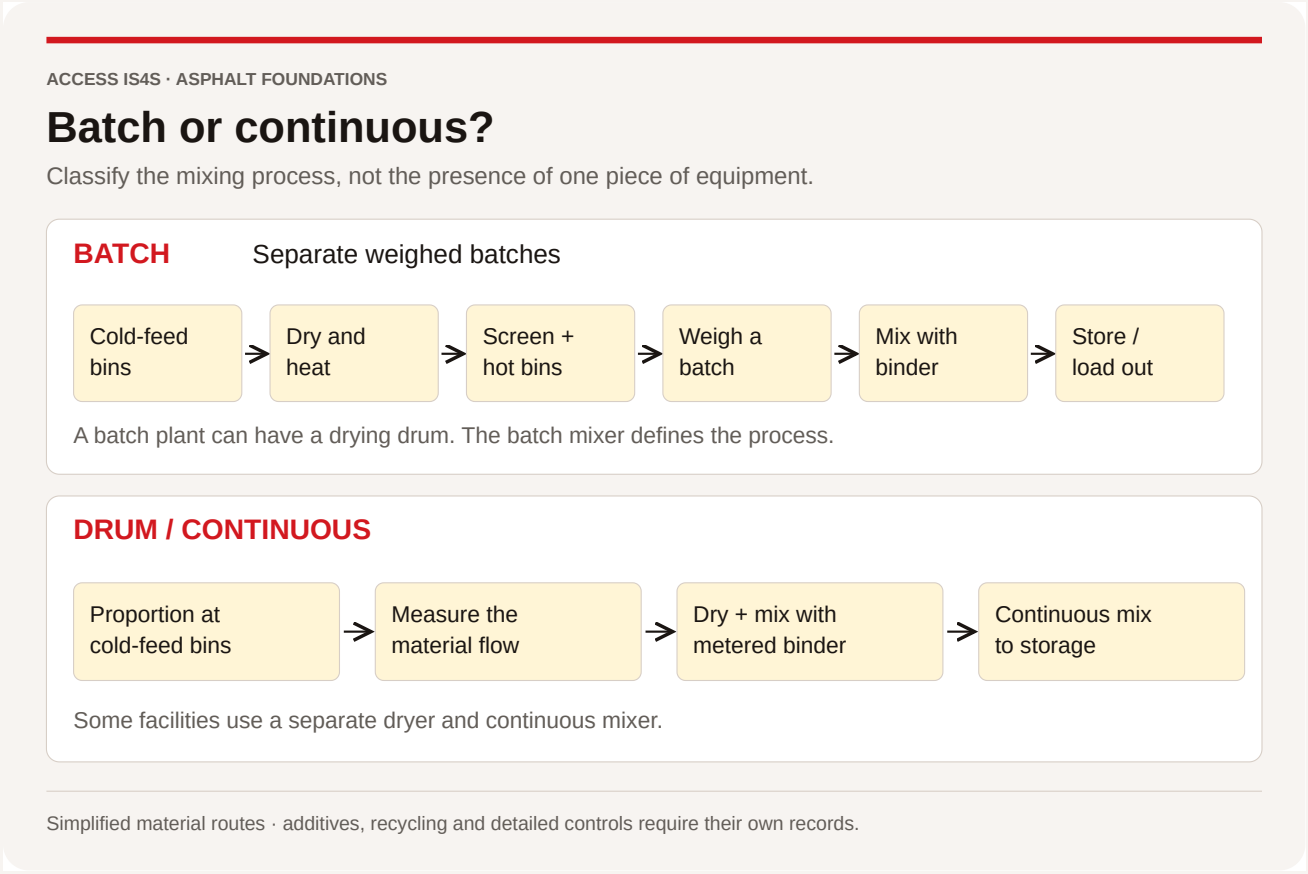
Write a three-sentence explanation that distinguishes aggregate, asphalt binder and fuel. Include the outline term *asphalt cement* and explain its relationship to *asphalt binder*. Add a sentence explaining the conditional purpose of an additive and why its use needs approved information. Distinguish a pit from a quarry, then name one question about an aggregate source that proximity alone cannot answer. Explain why “cement” needs context when comparing asphalt and concrete. **Skills:** Reading, Writing, Communication.

Distinguish batch and drum production

A **batch plant** prepares identifiable batches: aggregate is proportioned and combined with binder in a mixer. A **drum-mix plant** uses a continuous process in which material flows through the plant. Drum designs differ; do not treat every drum layout as identical. The beginner distinction is batch quantities versus a continuing material stream, not “one uses a drum and the other has no drum.” A batch plant can have a drying drum before its mixer.

Figure LA-4

Batch or continuous?



Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 7.2, p. 310 / PDF 344; Figure 7.4, p. 312 / PDF 346), by Asphalt Institute, 2007. Converted equipment drawings into two simplified process lanes; omitted settings and detailed control systems. Original Figure 7.2 is credited to Terex Roadbuilding in the handbook's contributor list. Original Figure 7.4 is credited to National Highway Institute (NHI) in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

Fictional classroom production records. Plant B's teaching log lists batch 01, batch 02 and batch 03, each with a separate record of ingredients and release. Plant D's teaching log records a continuous feed period and time-stamped checks. These are deliberately simplified descriptions. Neither log provides an approved mix design, production rate or operating instruction.

Worked example. “Plant B has a drying drum, so it must be a drum-mix plant” is an invalid inference. Ask where and how the mixture is produced. B's separate batch records support a batch process in this case. The physical presence of one part does not classify the whole system.

LA3-02 · Compare two process records

Identify the production type indicated by each fictional log. Draw or describe the common stages and the key difference. Add a correction to the worked-example claim in your own words. In pairs, one learner explains the map while the other checks whether it accidentally implies that a batch plant lacks a drying drum. Swap roles. **Skills:** Reading, Problem Solving, Collaboration.